

Prateek Singh

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EXPERIENCE

Evaratus (formerly Scaler AI Labs)

May 2026 – Present

Software Development Engineer (Intern)

Bangalore, India

- Build high-fidelity RL environments using Gymnasium and custom state-machine specs replicating live systems (web platforms, MCP servers, enterprise apps) as training and evaluation surfaces for frontier AI labs.
- Engineer evals, benchmarks and reward functions inside an applied-research loop of failure analysis, task design, verification and reward modelling; environments ship only after measured uplift on held-out tasks.
- Curate verified training data via end-to-end sourcing-to-delivery pipelines across regulated and consumer domains.

Google Summer of Code 2026 (writeup)

Nov 2025 – Present

Contributor, OWASP Foundation | OpenCRE

Remote

- Built Module C, the link decision engine inside OpenCRE, the open catalogue linking security standards (ISO, NIST, CWE, CAPEC) into one graph used by practitioners worldwide: candidate retrieval over pgvector, a cross-encoder reranker, and a decision layer gated on expected calibration error. **96.5% auto-link precision** at a 0.80 threshold and **98% retrieval recall** against a 319-chunk hand-labelled golden set.
- Designed it to escalate rather than guess: below-threshold candidates route to human review and the graph writer stays blocked until a safety detector exists. **247 hermetic tests** run in under 4s as a per-PR regression gate.
- Ran 13 reranker interventions that all regressed against the cosine baseline; root-caused it to 427 of 428 CREs having empty descriptions and published the negative result rather than ship an unvalidated fix.
- **2nd highest contributor** to the repository: 105 commits, 27,338 lines added. Link decisions now consumed downstream by OWASP Nest over MCP. Mentored incoming OWASP contributors through their first merged PRs.

Juni (writeup)

Apr 2026 – Present

Founder

India

- Solo-building a campus social app on a safety-first architecture: identity verification, consent-driven matching, ban-evasion tombstones, Safe Date mode with SOS dispatch. The safety layer was written before a single line of matching logic.

COMMUNITY AND TEACHING

Open Source Committee, Scaler School of Technology

2025 – Present

President (previously Core Member and Mentorship Lead)

Bangalore, India

- Lead the committee that gets students into open source from zero: Git and GitHub fundamentals, reading an unfamiliar codebase, finding a good first issue, and revising a pull request under maintainer review.
- Run build days and onboarding sessions; mentor 5+ students toward GSoC and other paid programs through contribution blockers, proposal writing and securing stipends.

PROJECTS

Sec2ndBrain (repo)

Tech: Next.js, TypeScript, Pinecone, Jina AI, Groq

- RAG knowledge base over personal content (notes, YouTube videos, tweets). Dual-path ingestion writes to MongoDB while embedding via Jina AI into Pinecone; queries are optimised by LLaMA 3.1 through Groq before retrieval.
- Kept MongoDB and Pinecone consistent against data drift and tuned the embedding-to-inference path for usable latency. End-to-end TypeScript.

Scaled Real-Time Chat (repo)

Tech: TypeScript, Node.js, ws, Redis Pub/Sub, Docker

- Room-based chat across parallel Node instances. In-memory Maps isolated rooms but only on one server, so I rearchitected it stateless around Redis Pub/Sub: servers publish to a channel and every instance delivers to its own WebSocket clients. Scales horizontally with no code change, demonstrated with two browsers on separate ports in one synchronised room.

EDUCATION

Scaler School of Technology

Aug 2024 – Aug 2028

Undergraduate Program in Computer Science

Bangalore, India

CGPA: 8.35

Birla Institute of Technology and Science (BITS), Pilani

Aug 2024 – Aug 2027

B.Sc. (Hons.) in Computer Science

Pilani, India

CGPA: 8.16

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, Go, Java, SQL. **ML & Retrieval:** RL environments (Gymnasium), eval harnesses, reward modelling, cross-encoder reranking, calibration, RAG. **Backend & Data:** Next.js, Express.js, React Native, three.js, MCP servers, PostgreSQL (pgvector), Pinecone, MongoDB. **Tools:** Git, Docker, GitHub Actions, pytest, hypothesis, ruff, mypy.

ACHIEVEMENTS

- **2nd highest contributor** to OpenCRE @ OWASP (Google Summer of Code 2026 @ OWASP Foundation).
- **CNCF** contributions: Headlamp, OpenKruise, Kuadrant mcp-gateway (PR #1141).
- **30% merit scholarship**, Algoverse AI Research Program, Summer 2026, from a global applicant pool.
- **500+ DSA problems** solved across LeetCode, Codeforces, CodeChef and Coding Ninjas.